Sharukesh M

B.Tech AI & Data Science Student

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Education

Amrita Vishwa Vidyapeetham — Coimbatore, India
B.Tech, Artificial Intelligence & Data Science | GPA: 8.78 — *Aug 2023 – Oct 2027*

Sri Chaitanya Techno School — Chennai, India
High School: 83.2% | Higher Secondary: 85.6% — *Apr 2021 – Apr 2023*

Experience

Telesto Energy — Singapore (Remote)
Artificial Intelligence Intern — *Dec 2025 – Mar 2026*

- Architected a scalable multi-modal RAG system using **AWS Bedrock**, **FAISS** for sub-millisecond vector search, and **Neo4j** for graph-based multi-hop retrieval.
- Leveraged the open-source **RAG-Anything** framework to ingest and index heterogeneous archival document corpora, enabling semantic search across thousands of records.
- Engineered an **OpenCV**-based document restoration pipeline to remove artifacts and sharpen degraded scans, significantly improving downstream OCR accuracy.

Discovery AI LLC — Plano, TX, USA (Remote)
Artificial Intelligence Intern — *Sep 2025 – Nov 2025*

- Built a web-based **AR Rug Visualizer** using **FastAPI** and **OpenCV** with floor plane estimation and projective geometry; mapped 2D product images onto room photos with accurate 3D perspective.
- Integrated **Pyodide** for browser-native Python execution, eliminating server round-trips and reducing visualisation latency for end users.

InceptAI — Coimbatore, India (Remote)
Artificial Intelligence Intern — *Mar 2025 – Apr 2025*

- Designed **Microsoft Security Copilot** plugins with **RAG pipelines** (DeepSeek R1) for context-aware KQL generation; implemented agentic workflows that measurably reduced SOC incident response time.
- Shipped plugins to production via **CI/CD pipelines** in collaboration with senior engineers.

Projects

Skin Lesion Classification — Jan – Apr 2025

- Designed a multi-task CNN (U-Net + ResNet) for skin lesion segmentation and classification (RCCN).
- Built a secure web/mobile app (FastAPI, PostgreSQL) for diagnosis, treatment recommendations, and geolocation-based appointment booking.

Scalable Academic Research Engine — Aug – Oct 2025

- Engineered a distributed data pipeline using **Apache Spark** (Scala) and **HDFS** to process and generate SciBERT embeddings for a massive corpus of academic papers.
- Deployed a high-throughput, **CUDA**-accelerated vector search backend using **FAISS** for real-time semantic similarity search.

Hemo-Flash: Efficient Hematology Detection — Jan – Mar 2026

- Developed **Hemo-Flash**, a lightweight **YOLOv11**-based system for real-time detection and classification of RBCs, WBCs, and platelets; achieved **86% reduction in model parameters** vs. SOTA YOLO.
- Engineered a high-performance inference pipeline for automated blood cell counting, balancing computational efficiency with State-of-the-Art diagnostic precision.

Publications

Mahendran, S. et al. "Analysis of Deep and Graph Neural Networks for Enhanced DDoS Attack Detection." *[IEEE, 2026].*

Mahendran, S. et al. "An Intelligent Plug-In Playable Orchestration Model Combining Gen-AI, RAG and Agentic AI for Cybersecurity Operations." *[Springer Nature, 2026].*

Technical Skills

Languages: Python, Java, C, SQL
AI / ML: PyTorch, TensorFlow, scikit-learn, FAISS, CUDA, Hugging Face Transformers, LangChain
LLM & RAG: RAG Pipelines, Agentic AI, Prompt Engineering
Computer Vision: OpenCV, Image Segmentation
Data & Systems: Hadoop, Neo4j (Graph DB), Redis, pandas, NumPy
Web & APIs: FastAPI, REST APIs, CI/CD Pipelines
Tools & Platforms: Git, Linux, AWS, Docker, Jupyter, VS Code
Functional skills: Problem solving, Team player